MATHEMATICAL ANALYSIS 1
HOMEWORK 5

(1) For each of the following functions, find all points $x_0 \in \mathbb{R}$ at which f is not defined or not continuous. At each such point compute $\lim_{x\to x_0^-} f(x)$ and $\lim_{x\to x_0^+} f(x)$, and classify the discontinuity (removable, jump, vertical asymptote, or of the second type). Explain your answers.
   (a) $f(x) = \begin{cases} x^2 & \text{for } x \neq 1 \\ 3 & \text{for } x = 1 \end{cases}$
   (b) $f(x) = \operatorname{sign}(x)$
   (c) $f(x) = \frac{x}{x-1}$, $\operatorname{dom}(f) = \mathbb{R} \setminus \{1\}$

(2) Let $f(x) = \lceil x \rceil$ be the ceiling function.
   (a) Prove that f is left-continuous at every $x_0 \in \mathbb{R}$.
   (b) Determine all points at which f is right-continuous, and classify the discontinuities of f.

(3) Prove Proposition 5.5 of the lecture notes:

Proposition. Let $f : \mathbb{R} \to \mathbb{R}$ be defined in a deleted neighborhood of x_0. Then

$$\lim_{x\to x_0} f(x) = L \qquad \Leftrightarrow \qquad \lim_{x\to x_0^+} f(x) = L \quad \text{and} \quad \lim_{x\to x_0^-} f(x) = L$$

where L can be any real number or $\pm\infty$. Moreover, if f is defined at x_0 it is continuous at x_0 if and only if it is both right- and left-continuous at x_0.

(4) Determine the values of $\alpha \in \mathbb{R}$ for which the following functions are continuous on their domains. Explain your answers.
   (a) $f(x) = \begin{cases} \alpha \sin(x + \frac{\pi}{2}) & \text{for } x > 0 \\ 2x^2 + 3 & \text{for } x \leq 0 \end{cases}$
   (b) $f(x) = \begin{cases} 3e^{\alpha x - 1} & \text{for } x \geq 1 \\ x + 2 & \text{for } x < 1 \end{cases}$

(5) Let $f(x) = \sin \frac{1}{x}$, with $\operatorname{dom}(f) = \mathbb{R} \setminus \{0\}$.
   (a) Find all $x \in (0, 1)$ for which $f(x) = 1$, and all $y \in (0, 1)$ for which $f(y) = -1$.
   (b) Using (a) and the definition of the right limit, prove that $\lim_{x\to 0^+} f(x)$ does not exist (it is neither a real number, nor $+\infty$, nor $-\infty$).
   (c) What type of discontinuity does f have at $x_0 = 0$?

(6) Let $x_0 \in \mathbb{R}$ and let f be a monotone increasing function defined on $\mathbb{R} \setminus \{x_0\}$ (it may or may not be defined at x_0). Following the proof of Theorem 5.6, define

$$L_- = \sup_{x<x_0} f(x) \qquad \text{and} \qquad L_+ = \inf_{x>x_0} f(x),$$

which are the left and right limits of f at x_0.
   (a) Explain why L_- and L_+ are real numbers (and not $\pm\infty$).
   (b) Prove that $L_- \leq L_+$. In particular, $L_+ < L_-$ is impossible.
   (c) Prove that if f is also defined at x_0, then $L_- \leq f(x_0) \leq L_+$ (this is Corollary 5.7).

(7) Prove the uniqueness of limits (Theorem 6.1) in the case of a finite point $x_0 \in \mathbb{R}$. There are two cases to prove: *(hint for both cases: by contradiction)*
   (a) If $\lim_{x\to x_0} f(x) = \ell_1$ and $\lim_{x\to x_0} f(x) = \ell_2$ with $\ell_1, \ell_2 \in \mathbb{R}$, then $\ell_1 = \ell_2$.
   (b) It is impossible to have both $\lim_{x\to x_0} f(x) = \ell \in \mathbb{R}$ and $\lim_{x\to x_0} f(x) = +\infty$.

(8) Let $f : \mathbb{R} \to \mathbb{R}$. Prove that if $\lim_{x\to+\infty} f(x) = \ell > 0$ or $\lim_{x\to+\infty} f(x) = +\infty$, then there exists $M > 0$ such that $f(x) > 0$ for all $x > M$ (this is the second item of Theorem 6.2). *Hint: the first item of Theorem 6.2 is proved in the lecture notes.*

(9) Let $f(x) = \frac{x^3-x^2+3}{x^2+3x+2}$.
   (a) Determine $\mathrm{dom}(f)$, and the set on which f is continuous.
   (b) At each real number that does not belong to $\mathrm{dom}(f)$, compute the left and right limits of f. What can you say about the graph of f near these points?

(10) Prove that
$$\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{1}{2}.$$
Hint: multiply both the numerator and the denominator by $1 + \cos x$, and use Example 5.13 of the lecture notes.

(11) In this problem we prove the **Squeeze Theorem** (Theorem 6.9) for a finite point x_0, and three functions f, g, h satisfying $f \le g \le h$ near x_0, with f and h having the limit $\ell \in \mathbb{R}$ as $x \to x_0$ (you are guided in the steps below):
   (a) State the theorem (and state that you shall prove it only in the case $x_0 \in \mathbb{R}$).
   (b) Fix $\varepsilon > 0$.
   (c) With this ε write the definition of what it means that $\lim_{x\to x_0} f(x) = \ell$.
   (d) With the same ε write the definition of what it means that $\lim_{x\to x_0} h(x) = \ell$.
   (e) Using the previous two steps, and the hypothesis $f \le g \le h$ near x_0, find a deleted neighborhood of x_0 (depending on ε) on which $|g(x) - \ell| < \varepsilon$.
   (f) Conclude that, since $\varepsilon > 0$ was arbitrary, the theorem follows by the definition of the limit (applied to g).

(12) Compute the following limits. In each case, explain which results from the lecture notes you use.
   (a) $\lim_{x\to+\infty} \frac{\cos x}{\sqrt{x}}$
   (b) $\lim_{x\to+\infty} \frac{\lfloor x \rfloor}{x}$
   (c) $\lim_{x\to 0} \frac{\sin^2 x}{x}$
   (d) $\lim_{x\to-\infty} x e^{\sin x}$
   (e) $\lim_{x\to+\infty} \left(x^2 + 3\sin x\right)$